RAJALAKSHMI ENGINEERING COLLEGE

B Keerthana 251801134-2025-AIDS

B2

| | |
|---|---|
| **Started on** | Thursday, 20 August 2026, 9:14 AM |
| **State** | Finished |
| **Completed on** | Thursday, 20 August 2026, 9:22 AM |
| **Time taken** | 8 mins 21 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

void func_with_counter(int n) {
    int count = 0;

    if (n == 1) {
        count++;
    } else {
        count++;
        for (int i = 1; i <= n; i++) {
            count++;
            for (int j = 1; j <= n; j++) {
                count++;
                count += 3;
                break;
            }
        }
        count++;
    }
    printf("%d\n", count);
}

int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        func_with_counter(n);
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course